Publication strategy and pre-submission risk register

Version 2026-11-01 (first version 2026-10-29), for Raz Vakil. Companion to `paper/clc.tex`.

CLC number: a squarefree composite n that is both Carmichael ((p−1) | (n−1) for every prime p | n) and Lucas–Carmichael ((p+1) | (n+1) for every p | n). The same objects appear in the literature as **1-Williams numbers** (see §5; the literature comparison is being completed and governs the final wording of every novelty claim).

1. What we have

#	result	type	status
1	**Theorem A.** A four-prime CLC number has all $p_i \equiv 1 \pmod{12}$, $p_3 < 2p_1p_2$, $p_4 < \sqrt{(2p_1p_2p_3)}$, $\mathbf{p_2^2 \le 24p_1^3 + 1}$, hence $n < 477\, p_1^{15/2}$; for each p_1 a finite search remains.	theorem, elementary	proof reconstructed twice; one error found and repaired in the AI audits; **awaiting human review**
2	**Theorem B.** No four-prime Williams number with $n \le 10^{30}$; none with $p_1 \le 500{,}000$ (any n). Extends McIntosh–Dipra's $p_1 > 2\times10^4$, $N > 10^{24}$.	computation	**certified** 2026-11-01 (`clc/k4/CERTIFY_*.txt`): full coverage, all survivors re-tested in Python
3	No CLC number below 10^{24} (factor-and-test of the complete Carmichael tabulations; completeness and per-class cross-checks; maximum of four matching primes).	computation on external data	done; a rescan recording the full match histogram is running (~21 h)
4	Synthesis of the Lucas–Carmichael / Baillie–PSW / Agrawal / Howe connections.	exposition	wording depends on §5
5	Polynomial-family impossibility theorems; exhaustive strata; subset-product margin measurement.	theorems + measurement	moved to Appendix A
6	Heuristics (four-prime expected count ≈ 6×10^{-7}; onset model).	speculative	labelled as such; onset model in Appendix B

Recommended core (per the external review of 2026-11-01): lead with Theorem A, the certified enumeration (Theorem B) and the 10^{24} scan; keep items 5–6 as appendices. Working title: *Explicit bounds and exhaustive searches for four-prime Williams numbers (Carmichael ∩ Lucas–Carmichael)*, to be fixed after §5.

2. Venues

The journals' AI policies decide eligibility; the assessments below are from the external review of 2026-11-01 and must be rechecked against the live policy pages immediately before submission.

venue	fit	eligibility / reservation	recommendation
arXiv (math.NT)	natural first step	permits disclosed AI assistance; a first-time submitter may need an endorsement	submit after Gates A–B
Canadian Mathematical Bulletin	good (Williams's question appeared there)	Cambridge requires declared AI use and full human accountability	**primary journal target**; ask the editor in advance about substantive AI involvement in proofs and code
Experimental Mathematics	good (computation + heuristics)	Taylor & Francis forbids generative-AI first drafts; detailed declaration required	possible only with a human-rewritten manuscript
Journal of Number Theory	stretch	plausible only if the inequality and the search advance are clearly substantial beyond McIntosh–Dipra (JNT 2015)	decide after §5
Mathematics of Computation	aspirational	needs algorithmic or certification novelty; endorsement from a consultant would matter	decide after §5
Journal of Integer Sequences	constrained	forbids LLM-written English; requires detailed credit for proof suggestions; discourages brute-force term extensions	only after a genuinely independent human rewrite
Integers	**ineligible**	its policy states it will not consider work using AI to produce mathematics, code or text	removed

OEIS: concise comments on A006972 / A002997 (and the near-miss data on A329948) only after the preprint identifier exists; OEIS is a record, not a priority mechanism. Code and owned data: Zenodo deposit with a versioned DOI; third-party inputs are cited and checksummed (`clc/DATA_SHA256.txt`), not redistributed.

3. Dissemination order

1. Finish the certification (Gate A), the literature comparison (§5) and the human proof review.
2. Send consultants the manuscript first, with assigned questions (proof correctness; enumeration completeness; prior art; venue level). Share the AI audit reports afterwards, so they do not anchor the first reading.
3. Freeze a release candidate: tag, archive code, logs, hashes and result files under a versioned DOI.
4. Submit a policy-compliant, human-accountable preprint to arXiv.
5. Post the OEIS comments, citing the arXiv identifier.
6. Address substantive feedback; submit to the chosen journal when the completion gates below are met, not after a fixed elapsed time.

4. Authorship and AI disclosure

The AI system (Claude, Anthropic) was used for proof discovery and drafting, code generation, prose drafting, literature search, and adversarial checking. The disclosure must name these roles separately (the current acknowledgement does). The author must independently reconstruct and accept responsibility for every proof, computation, citation and sentence; the two AI audit reports in `docs/` are adversarial checks, not independent human review, and are labelled accordingly. A consultant is offered co-authorship only for a contribution meeting ordinary authorship standards.

5. Literature and novelty

The primary-source check is written up in `docs/literature-williams-numbers.md`. Summary: the objects are **Williams numbers** (Echi 2007); McIntosh–Dipra (JNT 2015) proved finiteness with d–3 primes fixed, by the same product-of-quotients mechanism McIntosh used in 2014, and showed a four-prime example has $p_1 > 2\times10^4$ and $N > 10^{24}$; Zhang (Math. Comp. 2015) formalised the two-sided subset-product heuristics. The manuscript has been rewritten accordingly: name changed, prior results cited and compared, Theorem A presented as the explicit form of their finiteness theorem (its sharpness relative to their constants is **provisional** until Raz has read their PDF — free, link in the literature note), Theorem B as the extension from (2×10^4, 10^{24}) to (500,000, 10^{30}). Two TODO markers remain in the paper at exactly these two points. The all-k bound 10^{24} stands as the largest found in print, subject to the same read. Pinch's 10^{16} is stated with its parity restriction.

6. Risk register (neutral form)

risk	mitigation in place	still to do
Proof correctness (Theorem A)	two reconstructions; executable checks of every identity (`tests/test_theorems.py`); two AI audits, one of which found and repaired a false estimate in the $\Pi > 1$ case	human number-theorist review of §3
Enumeration completeness (Theorem B)	lemmas stated and proved; completeness proposition in §3; both candidate walks agree on six instances; relaxed-problem regression to 10^{12} archived; per-block logs; **certified** by `certify_k4.py` (`CERTIFY_*.txt`)	—
Verifier soundness	verifier checks the original tuple without normalisation; negative tests; C Fermat routine cross-checked against Python	—
Dependence on external tabulations (10^{24})	three completeness checks; per-ω cross-checks against two OEIS sequences; checksums	—
Prior art	literature note written; manuscript reframed	read McIntosh–Dipra and Zhang in full; resolve the two TODOs
AI policy compliance	roles disclosed; venue matrix	recheck policies at submission; human-authored prose where required

risk	mitigation in place	still to do
Process errors	the rerun's last blocks were lost to a `git stash -u` on live log files (2026-11-01) and recomputed; the wrapper now fails fast and certifies coverage	never run file-removing git commands while a job writes into the tree

7. Gates

Gate A — before any public posting - [ ] Theorem A proof text final (both Π branches explicit; δ sign correct); reviewed by a human number theorist. - [x] `CERTIFY_n1e30.txt` and `CERTIFY_p1_500000.txt` present, ending in CERTIFIED (3,811/3,811 and 323/323 blocks; 1,904 and 1,185 distinct survivors, all re-tested in Python) — 2026-11-01 13:04. - [x] Literature comparison written; novelty claims rewritten against McIntosh–Dipra and Zhang. - [ ] McIntosh–Dipra and Zhang read in full by the author; the two TODOs in the paper resolved. - [x] Implementation-completeness proposition written into §3 (Proposition 3.6).

Gate B — release candidate - [ ] `./run_tests.sh` passes from a clean clone on a second machine (transcript saved). - [ ] Source, compiler flags, binaries' hashes, inputs' hashes, command lines and logs frozen; Zenodo DOI. - [ ] 10^{24} histogram rescan finished, or the four-match count claim removed. - [ ] Every bibliography entry verified against the primary source.

Gate C — journal submission - [ ] Target journal's AI policy confirmed against the actual use history (written guidance if ambiguous). - [ ] Prose human-authored to the standard the venue requires. - [ ] Disclosure text matches the roles in §4. - [ ] Substantive consultant feedback resolved.